

Problem. Modern autonomous systems rely heavily on visual perception pipelines for tasks such as lane detection and trajectory estimation. While these systems often achieve strong per-frame accuracy, they are typically evaluated using static metrics like IoU or detection confidence, which do not capture how perception behaves over time. In practice, autonomous systems operate sequentially, so small perception errors can accumulate and propagate into unstable downstream behavior. This project investigates whether small, semantically plausible visual perturbations can induce hidden instability in autonomous visual perception systems before traditional confidence metrics indicate failure.

Rather than focusing on adversarial attacks, I study realistic environmental changes such as shadows, glare, faded lane markings, and partial occlusions. My central hypothesis is that these perturbations produce temporal and geometric instability disproportionate to their effect on per-frame accuracy: specifically, that instability metrics will increase substantially under perturbations that leave per-frame IoU within a small margin of the unperturbed baseline. I focus on lane perception and ego-trajectory estimation in driving scenes, analyzing how perception outputs evolve over time under perturbation and whether temporal inconsistency itself can act as a signal of uncertainty.

Proposed Methodology. This project will use the KITTI driving dataset and a classical lane perception pipeline to study counterfactual and temporal failure modes. I will build a baseline lane detection and trajectory estimation system using classical computer vision techniques (Canny edge detection, Hough transforms, and RANSAC-based lane fitting), with a lightweight pretrained model as a comparison only if time permits. The pipeline will estimate lane boundaries, centerline geometry, and approximate ego heading over video sequences. Before introducing perturbations, I will characterize baseline temporal jitter on clean sequences so that perturbation-induced instability can be meaningfully distinguished from inherent noise. I will then generate structured counterfactual perturbations simulating realistic driving conditions, focusing on three: partial lane fading, shadow and lighting shifts, and motion blur.

For each perturbed sequence, I will compare perception outputs against the original and measure temporal stability over time. Rather than evaluating only frame-level accuracy, I will introduce instability-based uncertainty metrics: lane boundary jitter across frames, vanishing point drift, trajectory oscillation, and accumulated heading deviation. The project connects directly to core CS 131 topics including image filtering, edge detection, geometric reasoning, camera models, and video understanding. My emphasis is not on improving detection performance, but on analyzing how uncertainty and instability emerge in autonomous perception systems under realistic conditions.

Next Steps. Week 1 will focus on KITTI dataset preparation, literature review, implementation of the classical baseline pipeline with trajectory visualization, and characterization of baseline jitter on unperturbed sequences. Week 2 will implement the perturbation framework and generate perturbed sequences under the three target conditions. Week 3 will focus on developing temporal stability metrics and running experiments across the perturbation settings. Week 4 will focus on quantitative analysis, visualization design, failure case analysis, and final demo and report preparation. The project is feasible within four weeks by relying on classical techniques aligned with CS 131 content and focusing on analysis and evaluation rather than large-scale model training.